

Graph transformer for molecular graph classification

In the four videos, we discussed transformers in regular domains such as text and images, where the input is naturally represented as a sequence of tokens or a grid of patches. In this assignment, we move to an irregular domain, where the input consists of a collection of labeled graphs

$$(G_1, y_1), \dots, (G_N, y_N),$$

where each graph is one sample. Each graph contains:

$$G_i = (V_i, E_i, X_i, Z_i, y_i),$$

where V_i is the set of nodes, E_i is the set of edges, X_i contains node features, Z_i contains edge features, and y_i is the graph-level label. The goal is to train a transformer-based model to predict the label of each graph.

Part A: Architecture design

Design a transformer-style architecture for graph-level prediction. Your description should address the following questions:

- Tokenization: here, we recommend using the graph nodes as tokens.
- Token embedding: How are the node features used to create initial token embeddings? Note that the original features are categorical. Treating them as standard numeric features will most likely yield suboptimal results.
- How do you address graphs of different number of nodes?
- How does the model use the graph structure? (Adjacency matrix, you can also use the edge features)
- How will you implement positional encoding (fixed? trainable?)
- Describe the attention blocks used to compute a representation for each token
- Describe the prediction head used for graph-level classification

Include a schematic description of the architecture.

Remark on computation If you don't have access to a GPU, I suggest design a transformer with a relatively small number of trainable parameters (smaller than 10^5).

Part B: Application to a graph dataset

Train your model on the ogbg-molhiv dataset.

This is a molecular graph classification dataset. Each graph represents a molecule. Nodes correspond to atoms, and edges correspond to chemical bonds. The task is binary classification: predict whether the molecule inhibits HIV replication. The dataset contains about 41,000 molecular graphs. Each node has 9 atom-level features, and each edge has 3 bond-level features. The dataset comes with predefined train/validation/test splits.

Results: add a plot with train and validation loss/accuracy. Select the best epoch according to validation ROC-AUC, and then report the corresponding test ROC-AUC.

Use of large language models (LLM) within the project

For this assignment - you can use LLM to produce the code according to your design. If you use LLM, attach the prompts used for generating the code. You are expected to understand the code generated by the LLM.